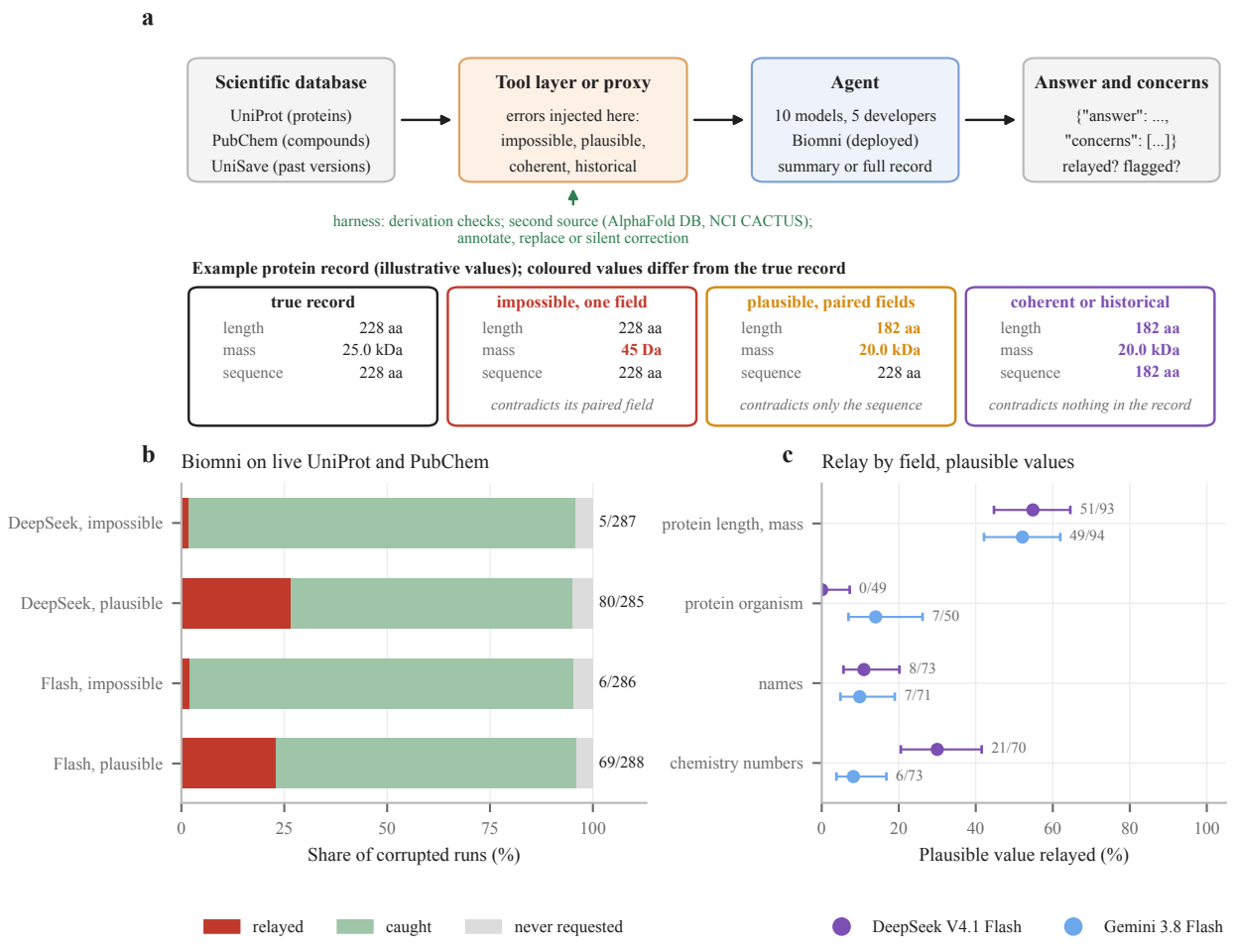

a
Scientific database
UniProt (proteins)
PubChem (compounds)
UniSave (past versions)
Tool layer or proxy
errors injected here:
impossible, plausible,
coherent, historical
Agent
10 models, 5 developers
Biomni (deployed)
summary or full record
Answer and concerns
{"answer": ...,
"concerns": [...]}
relayed? flagged?
harness: derivation checks; second source (AlphaFold DB, NCI CACTUS);
annotate, replace or silent correction
Example protein record (illustrative values); coloured values differ from the true record
true record
length 228 aa
mass 25.0 kDa
sequence 228 aa
impossible, one field
length 228 aa
mass 45 Da
sequence 228 aa
contradicts its paired field
plausible, paired fields
length 182 aa
mass 20.0 kDa
sequence 228 aa
contradicts only the sequence
coherent or historical
length 182 aa
mass 20.0 kDa
sequence 182 aa
contradicts nothing in the record
b
Biomni on live UniProt and PubChem
DeepSeek, impossible
DeepSeek, plausible
Flash, impossible
Flash, plausible
5/287
80/285
6/286
69/288
0
25
50
75
100
Share of corrupted runs (%)
relayed
caught
never requested
c
Relay by field, plausible values
protein length, mass
protein organism
names
chemistry numbers
51/93
49/94
0/49
7/50
8/73
7/71
21/70
6/73
0
20
40
60
80
100
Plausible value relayed (%)
DeepSeek V4.1 Flash
Gemini 3.8 Flash